



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

H2 BIOLOGY

9648/03

Applications Paper and Planning Question

26 September 2013

Paper 3

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your centre number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten ruled paper for
Question **5** separately.

Hand in the Question Paper, answers to Question **5**
separately.

The number of marks is given in brackets [] at the end of
each question or part question.

For Examiner's Use	
1	/ 15
2	/ 12
3	/ 13
4 (SPA)	/ 12
5	/ 20
Total	

This Question Paper consists of **17** printed pages.

Answer **all** questions.

1. DNA fingerprinting is a method of identification based on an individual's DNA. In traditional DNA fingerprinting, DNA is first isolated from cells, and then amplified by Polymerase Chain Reaction (PCR). **Fig. 1.1** illustrates the process of PCR.

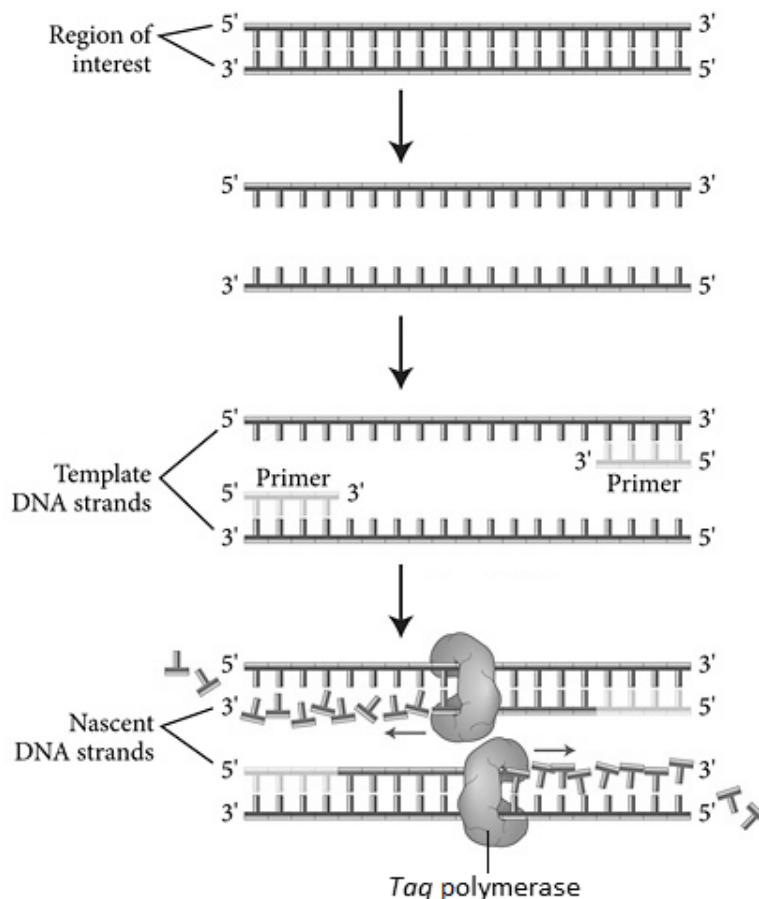


Fig. 1.1

- (a) (i) *Taq* polymerase is derived from *Thermus aquaticus*, a species of bacteria that thrives in hot springs. Explain why *Taq* polymerase is able to increase the amount of DNA from most species of organisms. [2]

1. Most species of organisms share the same genetic code
2. containing adenine, thymine, guanine and cytosine
3. Thus their DNA nucleotide can fit into the active site of *Taq* polymerase
4. for formation of phosphoester bonds between adjacent nucleotides

1. (a) (ii) Explain how the specific nucleotide sequence of the primer is important for its role in PCR. [2]

1. Allows for selective amplification
2. of a segment of DNA
3. via complementary base pairing
4. to 3' ends of the segment of interest

- (b) Distinguish between PCR and DNA replication in human skin cells. [2]

Factor	PCR	DNA
Nature of primer	DNA primer	RNA primer
Enzyme for polymerisation	<i>Taq</i> polymerase	<u>Human</u> DNA polymerase
Temperature	High temperatures between 60°C to 96°C	Body temperature of 37°C
Separation of template strand	Heat	Helicase

1. DNA amplified using PCR is then cut by a specific restriction enzyme and separated using gel electrophoresis. The DNA is then transferred to a nitrocellulose membrane, before the addition of radioactive DNA probes for nucleic acid hybridisation. **Fig. 1.2** shows the nucleic acid hybridisation process.

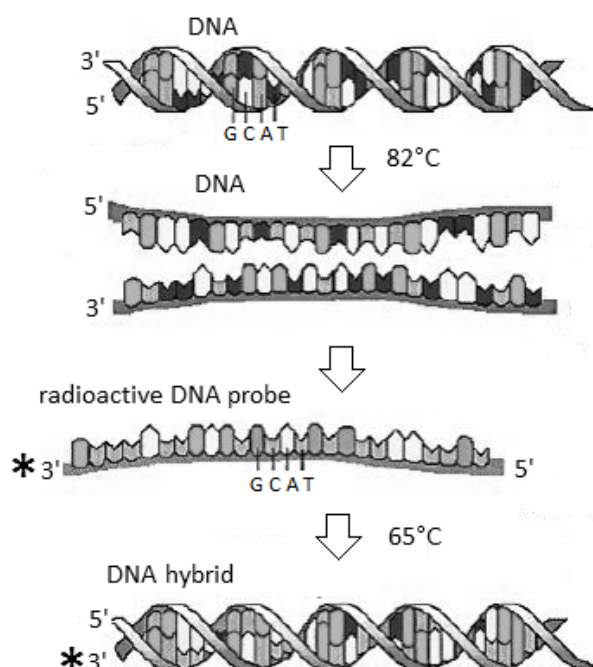


Fig. 1.2

1. (c) With reference to **Fig. 1.2**, explain the significance of the temperatures used during nucleic acid hybridisation.

[3]

82°C

1. Separation of DNA strands
2. by breaking hydrogen bonds
3. between complementary base pairs

65°C

4. Annealing of radioactive DNA probe
5. to DNA fragment of interest
6. via complementary base pairing
7. between GCAT and CGTA

1. (d) Suggest why several probes have to be used in order for DNA fingerprinting to be effective. [2]

1. To examine more than one VNTR locus
2. To obtain a combination of band patterns for comparison

DNA fingerprinting is often used in paternity testing. This involves the investigation of various Variable Number of Tandem Repeat (VNTR) loci in the individual's genome. The table below shows the genotype of five individuals at each of the five loci examined.

	Locus on chromosome				
	D19S433	D3S1358	SE33	D8S1179	D21S11
Mother	16,16	15,16	20,22	13,14	28,30
Child	16,18	16,16	22,24	13,14	28,29
Male 1	16,18	16,16	22,23	14,15	28,31
Male 2	17,18	15,16	24,25	11,12	28,30
Male 3	15,18	16,16	19,24	12,13	29,31

- (e) Define *Variable Number of Tandem Repeat*. [2]

1. DNA sequences
2. repeated in tandem
3. a variable number of times
4. at certain loci

- (f) (i) State the male(s) who may possibly be the father of the child. [1]

Male 3

Reject multiple answers

1. (f) (ii) Explain why the test above cannot conclude with absolute certainty that a given male is the father of the child. [1]

Not all VNTRs in the genome are verified as a match

Reject reference to coincidence without elaboration

[Total: 15]

2. In humans, the gene *RPE65* encodes a protein responsible for regenerating visual pigment in rod and cone cells after they have been exposed to light. A recessive allele of this gene causes impaired vision from birth, progressing to complete blindness in early adulthood. This condition is called LCA.

In 2008, trials were carried out into the possibility and safety of treating LCA using gene therapy.

- (a) Suggest and explain why LCA is suitable for treatment using gene therapy. [3]

1. caused by a single gene
2. caused by a recessive allele
3. delivery of, correct / dominant / normal, allele (could correct the condition)
4. ease of access to affected area
5. serious so worth the risk
6. only targets eye / no surgery needed
7. AVP

Six adults with this condition were used in the study. Genetically modified adenoviruses (a type of virus that can cause respiratory infections) were used as vectors. The vectors were injected beneath the retina of one eye of each of the participants.

- (b) Suggest two ways in which the genome of the adenoviruses used as vectors would differ from that of normal adenoviruses. [2]

1. virus no longer able to cause infections
2. correct / dominant / normal, allele (of RPE65) added
3. promoter added

Improvements were found in the vision of all the participants, but the small number in the trials made most of these improvements not statistically significant.

- (c) Suggest why these trials were designed to include such a small number of participants. [2]

1. ref. to safety / not known if the technique might have side effects
2. rare condition
3. expense
4. AVP e.g. trial to see if delivery method works

2. Scientists have developed stem cell therapy for retinal disease in mouse. The adult stem cells from donors form functional photoreceptors after transplantation in the adult diseased mouse retina. The immune system of the mouse was suppressed to prevent them from rejecting the foreign stem cells.

(d) Adult stem cells such as these are described as multipotent.

Explain what is meant by the term *multipotent*.

[1]

Multipotent cells can differentiate into a limited number of cell types (but not into whole organism)

(e) Suggest how it is possible that adult stem cells could differentiate into the range of cell types needed for repairing retina.

[2]

1. each stem cell contains a complete genome
2. cells will be in slightly different conditions / differentiate under appropriate conditions
3. through selective changes in genetic activity / different genes will be expressed, in different conditions

(f) Suggest the advantages of using stem cells from donors as compared to the use of adenoviruses in (b).

[2]

1. not all of the cells in the retina has taken up the normal allele
2. while all the stem cells from a donor will contain the normal allele
3. viral vector used to introduce the normal allele elicit host immune response
4. resulting in destruction of the vector

[Total: 12]

3. Rice is the staple diet in many parts of the world. Rice plants produce the precursor of vitamin A, β -carotene, in their green tissues, but not in the edible endosperms of their seeds. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease.

Golden rice, which contains β -carotene, was developed in Switzerland by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium. The rice produces seeds containing β -carotene.

Fig. 3.1 shows the DNA sequence used.



Key

pro – start site for polymerase enzymes

ter – end signal for polymerase enzymes

psy – gene from daffodil (a flowering plant)

crt 1 – gene from bacterium

Hyg resist – antibiotic resistance gene from a bacterium

Fig. 3.1

The following steps were taken to produce golden rice:

- Step 1:** a length of DNA (shown in **Fig. 3.1**) was made including a rice endosperm-specific promoter, *psy* gene, *crt 1* gene and *Hyg resist* gene,
- Step 2:** copies of this length of DNA were inserted into plasmids from the bacterium *Agrobacterium tumefaciens*,
- Step 3:** *A. tumefaciens* containing the plasmids were mixed with rice embryos and cultured in a medium containing the antibiotic *Hyg*,
- Step 4:** the embryos growing successfully on *Hyg* were grown into plantlets and then plants.

- (a) Describe the role of the rice endosperm-specific promoter that was added to *psy* and *crt 1* in **Step 1**. [2]

1. role of promoter is to switch on genes
2. ref to, transcription
3. only in the rice endosperm
4. not expressed anywhere else (would be a waste of energy / resources)
5. in edible part of seed

3. (b) Explain how a length of DNA can be inserted into a plasmid in **Step 2**. [3]

1. use restriction, enzyme / endonuclease
2. cut plasmid / open plasmid and the length of DNA
3. sticky ends / complementary bases
4. ref to anneal via hydrogen bonding / A - T / C - G
5. role of ligase, in sealing sugar-phosphate backbone / catalyses formation of phosphoester bond / AW

(c) Explain the role of the *Hyg* resistance gene in this procedure. [1]

1. To enable selection of the transgenic cells / cells that have taken up the plasmid
2. only the cells with the new DNA can grow in the presence of the antibiotic

Field trials of genetically engineered crops, such as golden rice, hope to identify any risks to the environment or to human health of growing and eating the crops.

(d) Suggest:

- (i) one possible risk to the environment of growing a genetically engineered crop; [1]

gene transfer to other rice

contamination of other crop (e.g. ref to organic crop)

gene transfer to wild relative

recipient plant outcompetes others / ref to 'superweed'

toxic to non pests

AVP must be something specific not a vague answer

any one

3. (d) (ii) one possible risk to human health of eating a genetically engineered crop.

[1]

allergy

long term toxicity

antibiotic resistance of gut bacteria if antibiotic marker used

gene transfer to, virus to human cells / gut bacteria, with unknown effect

AVP

any one

A group of adult volunteers were given a 500 g of labelled golden rice dose for three days together with 10 g butter. Each gram of golden rice contains 1.6 μg of β -carotene. Serum samples from the volunteers were analysed for golden rice β -carotene. The result of the analysis was shown in **Fig. 3.2**.

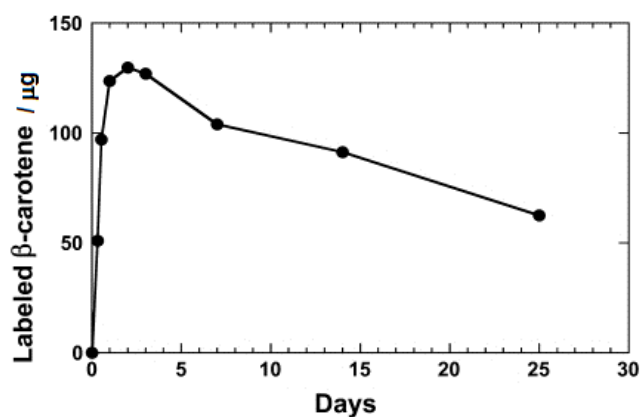


Fig. 3.2

- (e) With reference to **Fig. 3.2**, explain the change in labelled β -carotene.

[2]

- (i) Increase from day 0 to 3; due to intake

Decrease from day 3 to 25; due to use of β -carotene in vitamin A synthesis

- (ii) Suggest why the volunteers need to consume 10 g of butter together with the labelled golden rice.

[1]

β -carotene is fat soluble; facilitate intestinal uptake

3. (e) (iii) Suggest why children are not included in the study. [1]

Frequent blood draws are not suitable for children

- (f) The daily recommended intake of β -carotene is 2 mg.
Suggest if the daily intake of golden rice by children will overcome the problem of vitamin A deficiency. Explain your answer. [1]

No

Children consume less than 150 grams of rice a day/ less than adult

[Total: 13]

4. Planning Question

You are required to plan, but **not** carry out, an investigation into the effect of the independent variable, copper sulphate concentration, on amylase.

Amylase catalyses the hydrolysis of starch to glucose.

Copper sulfate can inhibit amylase.

The extent of inhibition depends on the concentration of the copper sulfate solution. A student investigated the inhibition of amylase at concentrations of copper sulfate solution greater than 3.0% and found that the enzyme was completely inhibited.

The student suggested the hypothesis:

Concentrations of copper sulfate solution below 3.0% will continue to inhibit the enzyme.

You are required to investigate this hypothesis by carrying out a serial dilution of copper sulfate solution which reduces the concentration by ten-fold between each successive dilution.

Using this information and your own knowledge, design an experiment to test the student's hypothesis.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- 1% amylase solution
- 1% starch suspension
- copper sulfate (powder)
- iodine in potassium iodide solution
- any normal laboratory glassware, e.g. test-tubes, beakers, measuring cylinders, glass rod, etc.,
- syringes,
- timer e.g. stopwatch,
- colourimeter,
- cuvette.

[Total: 12]

Introduction i.e. explanation of the dependent and independent variables involved [I=max 2]:

1. Amylase catalyses the hydrolysis of starch into glucose
2. Amylase has a specific active site
3. Starch fits into the active site to form an E-S complex
4. Resulting in the breakage of the glycosidic bonds
5. Copper sulfate inhibit amylase, prevent the binding of starch to the active site
6. The concentration of starch left can be measured by addition of iodine reagent
7. Measuring the colour intensity using a colourimeter
8. The intensity is proportional to the concentration of starch left and
9. inversely proportional to the rate of hydrolysis

Note: No mark for wrong concept

Expected trend [T=max 2]

10. As concentration of copper sulfate increases from 0% to 3.0%, the rate decreases
11. More enzymes are inactivated
12. Less enzyme-substrate complexes formed
13. Increased colour intensity
14. Beyond 3.0% copper sulfate
15. All the enzymes are inhibited
16. No reaction occurs

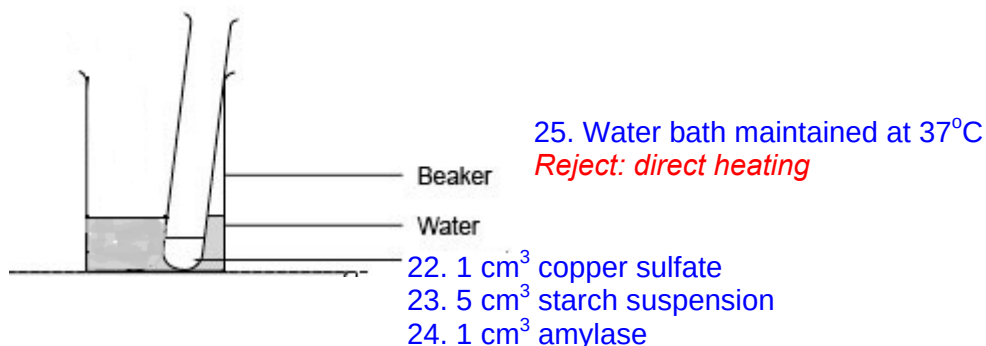
Experimental variables [V=max 2]

17. Independent variable : Concentration of copper sulfate
Range: 0.0003% to 3.0%
18. Dependent variable : Colour intensity in term of absorbance
19. Controlled variables :

	Controlled Variable	Appropriate Quantity
a	volume of starch suspension used	5 cm ³
b	volume of amylase used	1 cm ³
c	volume of copper sulfate used	1 cm ³
d	temperature	37°C / optimum temperature i.e. <u>equilibrate</u> amylase & starch to 37 °C separately <u>using water bath for 1 minute</u>
e	volume of iodine reagent added	0.5 cm ³

Method i.e. description of the method used including the scientific reasoning behind the method / relevant, clearly labelled diagrams: [M=max 3]

20. Preparation of 3.0% copper sulfate solution: dissolve 3.0 g of copper sulfate in 100 cm³ of distilled water.
21. Dilute the copper sulphate stock solution to give 10 cm³ each of the following concentrations: 0.0003%, 0.003%, 0.03% and 0.3%.
22. Add 1 cm³ of 3.0% copper sulfate into the test tube labelled as "3.0%"
23. Put 5 cm³ of starch suspension into the same test-tube.
24. Put 1 cm³ of amylase into the same test-tube.
25. Place the test tube in a water bath maintained at 37 °C.
26. Mix and start the stopwatch.
27. At the end of 10 minutes, add 0.5 cm³ of iodine reagent and mix well.
28. Transfer 1 cm³ of the mixture to the cuvette and measure the colour intensity using the colourimeter.
29. Record the reading in an appropriate table.
30. Repeat step 23 – 30 for the rest of copper sulfate concentration 0.0003%, 0.003%, 0.03%, 0.3% and control.



31. ensure that there are 3 replicates for each concentration of copper sulfate
32. Repeat the entire experiment twice times

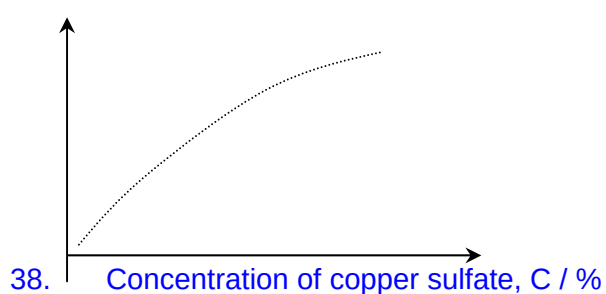
Control setup [C=max 1]

33. A negative control is subjected to the same factors (i.e. 5 cm³ starch suspension and 1 cm³ amylase) as that for the experiment, except the copper sulphate solution is replaced with equivolume of distilled water.
34. It is expected that yellow colour is observed upon addition of iodine reagent at the end of incubation time. This proves that it is indeed copper sulfate that inhibits the amylase

Data manipulation i.e. Include layout of results tables and graphs with clear headings and labels [D=max 2]

35. Concentration of copper sulfate, C / %	36. Colour intensity/ Absorbance			
	A_1	A_2	A_3	$\bar{A}$
0.0003				
0.003				
0.03				
0.3				
3.0				
control				

37. Absorbance



Safety precautions i.e. relevant risks and precautions taken: [S=max 1]

- 39. Amylase and iodine reagent are skin irritant. Wear glove and goggles
- 40. Copper sulfate is a harmful irritant. Wear gloves and goggles
- 41. AVP

Free Response Question

Write your answers on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

5. **(a)** Distinguish between a genomic DNA library and cDNA library, and outline their formation. [8]

Formation

gDNA

1. Entire genome is cleaved
2. with a specific restriction endonuclease
3. Resulting fragments are inserted into plasmids before introducing into bacteria

cDNA

4. Total RNA
5. converted into single-stranded cDNA
6. Using reverse transcriptase
7. Resulting cDNA molecules are inserted into plasmids before introducing into bacteria

Differences

	Feature	Genomic DNA library	cDNA library
8.	DNA fragments in clones	Contains DNA fragments representing the entire genome	Contains DNA fragments representing only the coding region of a genome
9.	Introns	Introns are included in the clones	No introns present in the clones (intron sequences removed by RNA splicing)
8.	Frequency	All genes are represented equally in the library	Genes that are transcribed abundantly will be represented more frequently in the library
9.	Number of clones	Larger number of clones to screen	Smaller no. of clones to screen (as it contains only the coding regions)
10.	Regulatory sequences	Presence of regulatory sequences	Absence of regulatory sequences
12.	AVP		

(b) *“Gene therapy clinical trials have mostly shown to be unsuccessful in terms of therapeutic interest.” Explain why.* [7]

1. Difficult to get DNA to integrate into the target cell genome
2. Results can be short-lived / patients need to be treated on a frequent basis
3. Problem with controlling the activity of gene expression
4. Risk of stimulating immune response
5. Virus vector may regain / develop virulence
6. Incorrect insertion of gene may cause named / described problem to patient
7. Possible problem with finding vector for large gene
8. Many genetic diseases are a result of presence of many defective genes / impossible to introduce many genes at same time
9. Problem dealing with, dominant condition / non dividing cells
10. Toxicity / low efficiency of liposomes
11. AVP

(c) The genetic manipulation of plants and animals, and their use in agriculture, is one the most controversial scientific development of recent times. What are some moral issues associated with the rise of such technology? [5]

1. Violation of natural organism's intrinsic values with the mixing of genes among species / tampering of genes
2. Animal welfare could be adversely affected due to little concern about whether animals could withstand the additional burden of increased milk / meat production
3. Controversies on patent of GMO as patent of transgenic plants / animals reduces them to the level of objects
4. Religious / dietary concerns with consuming animal genes in plants / relevant examples
5. Labelling of GM food is not mandatory in some countries thus lack of awareness for those who feel strongly against consumption of foreign DNA incorporated into their food
6. AVP

[Total: 20]